Journal of Energy Storage 101 (2024) 113899

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Research papers

A three-dimensional multiphysics field coupled phase field model for lithium dendrite growth

Guoqing Qi [a], Xunliang Liu [a,b,*], Ruifeng Dou [a,b], Zhi Wen [a,b], Wenning Zhou [a,b], Lin Liu [c]

[a] School of Energy and Environmental Engineering, University of Science and Technology Beijing, Beijing 100083, China
[b] Beijing Key Laboratory of Energy Saving and Emission Reduction for Metallurgical Industry, School of Energy and Environmental Engineering, University of Science and Technology Beijing, Beijing 100083, China
[c] Department of Mechanical Engineering, The University of Kansas, Lawrence, Kansas 66045, USA

ARTICLE INFO

Keywords:
Lithium battery
Lithium dendrite
Three-dimensional phase field
Morphological simulation

ABSTRACT

Lithium metal is considered an ideal anode material for future lithium-ion battery technology due to its high energy density and low redox potential. However, the growth of lithium dendrites can lead to separator membrane penetration, resulting in internal short circuits and, ultimately, battery failure. This study developed a three-dimensional multiphysics phase field model to thoroughly examine the growth mechanism of lithium dendrites and investigate more effective methods for their inhibition. We successfully achieved three-dimensional simulations of the growth of various forms of lithium dendrites, such as dendritic, mossy, needle-like, and multi-branching crystals, and compared their different growth characteristics. Additionally, we analyzed the effects of both anisotropy and temperature on lithium dendrite growth by coupling temperature field analysis. Finally, combining the analysis of internal potentials in multi-branching crystals, we've proposed a possible approach to enable stable lithium deposition processes. These research findings can provide essential theoretical support for further development in lithium metal batteries.

1. Introduction

With the rapid development in consumer electronics, electric vehicles, and chemical energy storage, demand is increasing for higher energy density and battery safety [1]. Compared to traditional graphite anodes, lithium metal anodes possess an exceptionally high theoretical energy density, making them the 'holy grail' in the battery domain [2–5]. However, during the charge-discharge processes of lithium metal batteries, dendrite growth can lead to membrane penetration, internal short circuits, and the formation of 'dead lithium,' issues that have long hindered the further advancement of lithium metal batteries [2,3,6–8]. Therefore, gaining a deep understanding of dendrite growth and 'dead lithium' formation mechanisms and exploring methods to facilitate stable deposition and decomposition of lithium metal holds significant importance for the commercial application of lithium metal batteries.

Meanwhile, the phase field method has shown unique advantages in understanding lithium dendrite evolution. This method utilizes the diffuse interface technique, which can easily track complex microstructural interface changes. The phase field model can be conveniently coupled with various multiphysics governing equations (such as temperature, fluid mechanics, and solid mechanics) to investigate the influence of different factors on dendrite evolution [9–11]. Since initial attempts to establish a two-dimensional phase field model for lithium dendrite growth [12,13], most simulation studies are still limited to two-dimensional space. Coupling phase-field simulations of lithium dendrite growth with a heat transfer model can reveal the impact of internal heat generation and environmental temperature on dendrite growth patterns [14,15]. Based on the thermally coupled phase field model, the effects of anisotropy, applied voltage, SEI microstructure, comprehensive pulse voltage, and temperature on dendrites have been effectively explored [16,17]. Additionally, after the introduction of solid electrolytes, the force-chemical-electric coupling behavior of lithium dendrite growth along grain boundaries and crack propagation has also received widespread attention [18–23,50,51]. Meanwhile, the role of the reaction between electrolyte and lithium metal to form a solid-state SEI layer on the growth of internal lithium dendrites was effectively explored by the phase-field approach of coupled stress and crack extension [24,25]. Structured lithium metal anode approaches have recently garnered significant interest in achieving safe and high-energy density lithium metal batteries. Experimental evidence has shown that a vertically

* Corresponding author at: School of Energy and Environmental Engineering, University of Science and Technology Beijing, Beijing 100083, China.
E-mail address: liuxl@me.ustb.edu.cn (X. Liu).

https://doi.org/10.1016/j.est.2024.113899
Received 7 July 2024; Received in revised form 2 September 2024; Accepted 20 September 2024
Available online 26 September 2024
2352-152X/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

aligned structure with minimal curvature in the negative electrode framework is favorable for lithium metal deposition [26]. In contrast, the growth process of lithium dendrites on vertically aligned frameworks has been simulated using the phase field method [27]. The impact of carbon paper fibers [28] and gradient porosity [29] on lithium deposition has also been preliminarily studied using the phase field method. However, studying lithium deposition behavior within three-dimensional porous frameworks using only two-dimensional models has significant limitations. In terms of ion transport, two-dimensional models cannot capture the transport of lithium ions from all directions within three-dimensional porous structures, potentially leading to less precise simulation results. Moreover, two-dimensional models cannot calculate the spatial potential distribution of complex framework structures, making it difficult to accurately investigate the role of charged three-dimensional frameworks in promoting uniform deposition and stable decomposition.

For the electrodeposition process on a practical scale, especially for the optimized design of the negative electrode porous skeleton, it is no longer sufficient to rely only on two-dimensional planar simulations. Despite the higher computational cost of simulating at a three-dimensional scale, three-dimensional simulation methods should be adopted to enhance our understanding of experimental phenomena. [30]. For example, the development of current collectors and anode frameworks in lithium metal batteries [26,31–35], artificial SEI layers [36–38], and the design of electrolyte ion channels [39,40] at real length scales have heightened the need for three-dimensional electrodeposition models. These models are crucial for elucidating the transport of lithium ions within porous structures, the spatial distribution of potential, and the deposition process of lithium ions within porous electrodes at a three-dimensional real length scale. Mu et al. [41] significantly reduced the computation time for three-dimensional lithium dendrites using the open-source software AMReX through parallel computing; however, their description of the methodology for establishing the 3D model remains insufficient to be implemented. Moreover, the simulation results are limited to multi-core growth of lithium dendrites, without adequate differentiation between primary and secondary branches. Arguello et al. [42,43] successfully simulated three-dimensional dendritic lithium growth using MOOSE by incorporating a fourfold anisotropy approach from alloy solidification; however, the complexity of their model is relatively high, and the results predominantly focus on dendritic morphologies, with less attention given to other dendrite shapes. Current research on the coupling of temperature fields in three-dimensional lithium dendrite models remains insufficient. At present, COMSOL remains the mainstream software for simulating lithium dendrite growth, yet relevant studies specifically addressing the three-dimensional lithium dendrite simulation have yet to be reported. Overall, research on three-dimensional modeling of lithium dendrites is still at an early stage, necessitating significant advancements in related models.

This study develops a three-dimensional phase field model for the growth of lithium dendrites by tuning anisotropy parameters. It simulates the formation processes of dendrites in various morphologies, including dendritic, mossy, needle-like, and multi-branched forms. It further evaluates and compares physical parameters such as growth rate, deposition ratio, and solid phase fraction across different morphological forms. By integrating temperature field coupling, this research investigates the maximum height of dendrites and the variations in solid phase fraction with changes in anisotropy and temperature at a three-dimensional scale. Additionally, an analysis of the three-dimensional potential within multi-core lithium dendrites suggests a potential strategy for achieving stable lithium metal deposition. This study demonstrates a viable method for simulating three-dimensional lithium dendrites using the phase field approach.

2. Mathematical model

2.1. Phase field model

Inspired by Chen [12] and Yan [14], the developed phase field method based on the principle of minimizing free energy, while introducing the phase field variable ξ *(x,y,z,t)* to characterize the physical state of each point in the system at a certain moment: $\xi = 1$ represents the solid-phase lithium dendrite, $\xi = 0$ represents the liquid-phase electrolyte, and the region where $0 < \xi < 1$ represents the solid-liquid phase interface. The phase field variable on the solid-liquid phase interface continuously changes from 1 to 0, thus avoiding directly tracking the solid-liquid phase interface. The model assumptions are summarized as follows: (1) Electrolytes follow the dilute solution theory; (2) The electron supply on the electrode surface is sufficient; (3) The electrode phase is metallic lithium, ignoring the effect of the SEI layer. The Gibbs free energy of the system can be expressed as:

$$F = \int_V \left[f_{ch}(c) + \rho_e \varphi + \frac{1}{2} \nabla C_a \cdot K_n \nabla C_a \right] dV \tag{1}$$

where $c = \{c_a, c_+, c_-\}$represent the concentrations of Li atoms, Li^+ ions, and PF_6^- ions, respectively; ρ_e and φ are the charge density and electrostatic potential, and their product represents the electrostatic energy density; K_n is the gradient energy coefficient; V is an arbitrary volume. Normalizing the concentrations dimensionless $\left\{ C_a = \frac{c_a}{c_s}, C_+ = \frac{c_+}{c_0}, C_- = \frac{c_-}{c_0} \right\}$, where c_s is the site density of Li metal, with a value of 7.64 $\times 10^4$ mol/m^3, and c_0 is the volume concentration of Li^+ ions, with a value of 1.0 $\times 10^3$ mol/m^3.

The first term $f_{ch}(c)$ in Eq. 1 is related to the chemical potential and can be expressed as [12,14]:

$$f_{ch} = c_s RTC_a lnC_a + c_0 RT(C_+ lnC_+ + C_- lnC_-) + \sum_i c_i \mu_i^{\ominus} \tag{2}$$

where $C_a = 1$ represents in the solid phase of lithium metal, $C_a = 0$ represents in the liquid phase of the electrolyte, to describe the diffusion interface in the phase field model, use the double-well potential $g(\xi) = W\xi^2(1-\xi)^2$ instead of $c_s RTC_a lnC_a$, where$\xi = C_a$, $W/16$ represents the barrier height, the free energy density of the electrochemical potential can be further written as:

$$f_{ch} = W\xi^2(1-\xi)^2 + c_0 RT(C_+ lnC_+ + C_- lnC_-) + \sum_i c_i \mu_i^{\ominus} \tag{3}$$

where R is the molar gas constant; $\mu_i^{\ominus}$ is the reference chemical potential of substance *i*.

By taking the variation of the total free energy with respect to ξ, we can obtain:

$$\frac{\partial \xi}{\partial t} = -M \frac{\delta F}{\delta \xi} = -M\left(g'(\xi) - K_n \nabla^2 \xi + H(\xi) \right) \tag{4}$$

The parameter $H(\xi)$ is related to the driving force of the electrochemical reaction, and M represents the interface mobility. The approximate expression for $H(\xi)$ can be derived from the electrochemical reaction kinetics.

For the $Li^+ + e^- = Li$ chemical reaction, the reaction rate v_e is described by the Butler-Volmer equation and can be expressed as follows [12]:

$$v_e = -k_0 a_+^{1-\beta} a_{Li}^{\beta} \left\{ exp\left[\frac{(1-\beta)nF\eta}{RT} \right] - exp\left[\frac{-\beta nF\eta}{RT} \right] \right\} \tag{5}$$

where k_0 is the reaction rate constant; $n$is the number of electrons exchanged in the reaction, taken as 1; β is the asymmetry factor, taken as 0.5; F is the Faraday constant, 96,485C/mol; R is the gas constant, 8.314

$J/(\mathrm{kg}\bullet\mathrm{K})$; and T is the temperature, taken as 298 K.

The overpotential η is defined as the sum of the activation overpotential η_a and the concentration overpotential η_c:

$$\eta = \eta_a + \eta_c = \Delta\phi - E^{\Theta} - \frac{RT}{nF} ln \frac{a_{\mathrm{Li}+} a^{-}}{a_{\mathrm{Li}}} \tag{6a}$$

which can be further expressed as:

$$\eta = \Delta\phi - E^{\Theta} - \frac{RT}{nF} ln \frac{C_{+}}{a_{\mathrm{Li}}} \tag{6b}$$

By substituting Eq. 6b into Eq. 5, the electrochemical reaction rate can be expressed as:

$$v_e = -k_0 \left\{ a_{\mathrm{Li}} exp\left[\frac{(1-\beta)nF\eta_a}{RT}\right] - C_{+} exp\left[\frac{-\beta nF\eta_a}{RT}\right] \right\} \tag{7}$$

where, a_{Li} is approximately treated as the phase field variable $\xi^{\frac{c_0}{c_s}}$, and further:

$$v_e = -k_0 \left\{ \xi^{\frac{c_0}{c_s}} exp\left[\frac{(1-\beta)nF\eta_a}{RT}\right] - C_{+} exp\left[\frac{-\beta nF\eta_a}{RT}\right] \right\} \tag{8}$$

Assuming the phase field evolution at the interface is proportional to the electrochemical driving force:

$$H(\xi) = -k_0 h'(\xi) \left\{ \xi^{\frac{c_0}{c_s}} exp\left[\frac{(1-\beta)nF\eta_a}{RT}\right] - C_{+} exp\left[\frac{-\beta nF\eta_a}{RT}\right] \right\} \tag{9}$$

where $h'(\xi)$ is the first derivative of the interpolation function $h(\xi) = \xi^3(10 - 15\xi + \xi^2)$, which constrains the electrochemical driving force acting only at the phase field interface. Therefore, the phase field Eq. 4 can be written as:

$$\tau \frac{\partial \xi}{\partial t} = K_{\mathrm{n}} \nabla^2 \xi - g'(\xi) - k_0 h'(\xi) \left\{ \xi^{\frac{c_0}{c_s}} exp\left[\frac{(1-\beta)nF\eta_a}{RT}\right] - C_{+} exp\left[\frac{-\beta nF\eta_a}{RT}\right] \right\} \tag{10}$$

where $\tau = 1/M$ is a small time constant that can be obtained experimentally. The dendritic morphology in three dimensions can be achieved by modifying the gradient energy coefficient [44]:

$$k = \lambda_0 [1 + \delta cos(\omega\theta)] \tag{11}$$

$$K_{\mathrm{n}} = \begin{pmatrix} k^2 & -kk' & -kk' \\ kk' & k^2 & -kk' \\ kk' & kk' & k^2 \end{pmatrix} \tag{12}$$

where $\lambda_0$is a pre-factor related to surface energy, $\delta$is the anisotropy strength, and $\omega$is the anisotropy modulus;

In the three-dimensional model, θ can be defined as the angle between the unit normal direction $\vec{n} = \frac{(\xi_x, \xi_y, \xi_z)}{\sqrt{\xi_x{}^2 + \xi_y{}^2 + \xi_z{}^2}}$ of the interface and the xoy plane, i.e., $\theta = arctan\left(\frac{\xi_z}{\sqrt{\xi_x{}^2 + \xi_y{}^2}}\right)$, as shown in Fig. 1.

Considering only the migration of Li^+ in the electrolyte, its driving force is the concentration gradient and the electric potential difference. The flux can be expressed as [12,14]:

$$J_i = -M_i c_i \nabla \mu_i = -D_i \left(\nabla c_i + c_i \nabla \frac{F\varphi}{RT} \right) \tag{13}$$

By neglecting the diffusion of lithium atoms and anions ($PF_6{}^-$) in the model, we can further obtain:

$$\frac{\partial C_{+}}{\partial t} = \nabla \left[D_{\mathrm{eff}} \nabla C_{+} + \frac{D_{\mathrm{eff}} C_{+}}{RT} nF \nabla \varphi \right] - \frac{c_{\mathrm{s}}}{c_0} \frac{\partial \xi}{\partial t} \tag{14}$$

where D_{eff} is the effective diffusion coefficient.

The last term on the right side of the Eq.14 represents the rate of production or consumption of Li^+ due to electrochemical reactions at the electrode surface. The effective diffusion coefficient can be expressed by the interpolation function:

$$D_{\mathrm{eff}} = D_{\mathrm{s}} h(\xi) + D_{\mathrm{l}} (1 - h(\xi)) \tag{15}$$

where D_{s} is the diffusion coefficient of Li^+ in the electrode, D_{l} is the diffusion coefficient of Li^+ in the electrolyte.

Assuming the system is electrically neutral, the charge conservation is described by the Poisson equation, with an added source term to represent the charge consumed or produced due to electrochemical re-

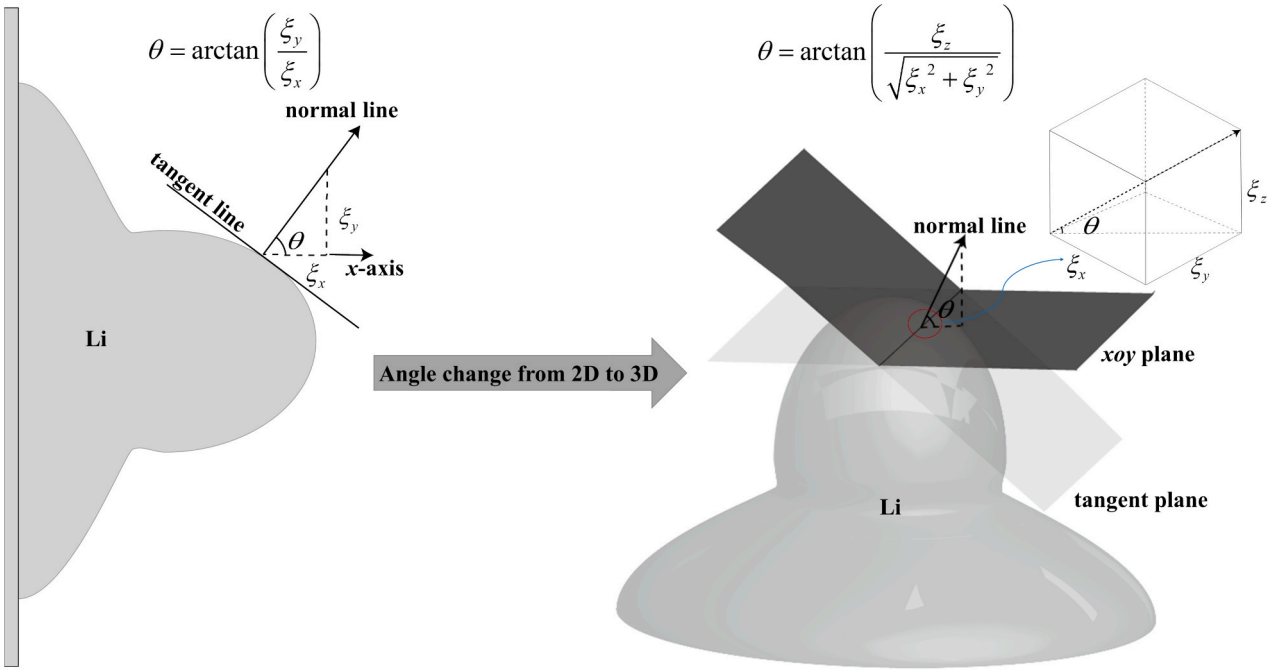


Fig. 1. Variation of anisotropic pinch angle from 2D model (left) to 3D model (right) expansion.

actions [12,14]:

$$\nabla \cdot \sigma_{\text{eff}} \nabla \varphi(r,t) = J_R \tag{16}$$

where σ_{eff} is the effective electrical conductivity; the source term $J_R = nFc_s\frac{\partial \xi}{\partial t}$ is generated only at the electrode-electrolyte interface when the system deviates from its equilibrium state.

Similar to the effective diffusion coefficient, the effective electrical conductivity is expressed as:

$$\sigma_{\text{eff}} = \sigma_s h(\xi) + \sigma_l(1 - h(\xi)) \tag{17}$$

where σ_s is the conductivity of the electrode, σ_l is the conductivity of the electrolyte.

The change in battery temperature is governed by the heat conduction differential equation:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot \lambda \nabla T + Q \tag{18}$$

where Q is the internal heat source term composed of Ohmic heating Q_{ohmic} and overpotential heating Q_{op}, which can be expressed by the following equation:

$$Q = Q_{\text{ohmic}} + Q_{\text{op}} = \sigma^{\text{eff}}(\nabla \varphi)^2 + a_s nFc_s \frac{\partial \xi}{\partial t}\varphi \tag{19}$$

a_s is an empirical factor with a value of 0.033 [15]. The effect of temperature on the lithium-ion diffusion rate can be expressed by the following empirical formula:

$$D^l = D_0 exp\left[m(1 - C_+) - \frac{E_d}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right] \tag{20}$$

where T_0 represents the ambient temperature, taken as 298 K, D_0 represents the diffusion coefficient of the electrolyte at 298 K, and $m = 1.3$ [17]is the correction factor.

2.2. Numerical calculation method

The three-dimensional phase field equations for lithium dendrite growth were solved using the finite element method in the Partial Differential Equation (PDE) module of Comsol Multiphysics 6.0. For enhanced computational efficiency, a cubic computational domain measuring 200 × 200 × 350 μm^3 was utilized. An initial nucleation point was positioned at the center of the bottom with an initial phase field order parameter value of 1, a concentration of 0, and a voltage of −0.1 V. At the top, the initial phase field order parameter value was set to 0, the concentration to 1, and the voltage to 0. Periodic boundary conditions were applied to the sides of the domain. Initial values within the domain were set as follows: phase field order parameter at 0, concentration at 1, and electric potential field at 0. Both convection and radiation boundary conditions were employed in the temperature field. The adaptive mesh refinement feature of Comsol was utilized. The simulation of dendrite growth over 160 s required approximately 100 min of computational time on a desktop computer equipped with an Intel® Xeon® Processor E3–1230 v2. Details of the model parameters are provided in Table 1 of the attachment.

3. Results and discussion

3.1. Three-dimensional simulation of lithium dendrites

Based on experimental studies of lithium dendrites in liquid batteries, the morphology of lithium dendrites can be classified into four types: dendritic [45], mossy [46], needle-like [47], and multi-core growth [48]. Noise terms are introduced into the phase field equation to simulate mossy dendrites and achieve their characteristic morphology. Needle-like dendrites are simulated by restricting the transport of lithium ions to achieve their morphological features. Multi-core growth lithium dendrites are achieved by introducing lithium sheets at the bottom and applying noise terms.

3.1.1. Three-dimensional morphology of lithium dendrites

Fig. 2 shows cross-sectional views of the phase, potential, and concentration fields for four types of three-dimensional lithium dendrites, along with the corresponding experimental results. The comparison reveals that the simulated cross-sectional results are consistent with the two-dimensional results provided by Chen et al. [12], Mu et al. [16], and Zhang et al. [49], and match the observed dendritic (Fig. a4) [45], mossy (Fig. c4) [46], needle-like (Fig. b4) [47], and multi-core growth (Fig. d4) [48] lithium dendrites. This consistency demonstrates the feasibility of the model. The temporal evolution of the phase field, potential field, and concentration field for the three-dimensional structures of lithium dendrites can be referenced in the attachments (S_1: three-dimensional dendritic; S_2: three-dimensional needle-like; S_3: three-dimensional mossy; S_4: three-dimensional multi-core growth lithium dendrites).

3.1.2. Three-dimensional lithium dendrite data analysis

In the phase field results, regions with $\xi > 0.05$ were selected to remove the electrolyte to make the three-dimensional morphology of the dendrites clearer.

Fig. 3(a) shows the dimensionless height variation curves of dendritic, needle-like, and moss-like dendrites over time. The results indicate that the growth rate of needle-like crystals in the height direction increases almost linearly and is significantly faster than that of dendritic and moss-like lithium dendrites. During 0–20 s, the growth rate of moss-like dendrites in the height direction is higher than that of dendritic ones, but subsequently, the growth rate in height gradually decreases and becomes less than that of dendritic dendrites. Fig. 3(b) presents the variation curves of the ratio of main branch deposition (S_M) to side branch deposition (S_L) over time for moss-like and dendritic crystals. In moss-like crystals, the main branch deposition (S_M) is defined as the deposition within a cylindrical volume with a radius equal to half the comprehensive radius of the base, while the side branch deposition (S_L) is defined as the deposition outside this cylindrical volume. From Fig. 3 (b), it can be concluded that the deposition in dendritic crystals is concentrated in the main branch, whereas in moss-like crystals, the deposition is concentrated in the side branches. Fig. 3(c) shows the variation of the ratio of main branch height (b) to side branch length (a) over time for needle-like and dendritic dendrites. The b/a ratio for needle-like crystals approximates a straight line, indicating a continuously accelerating growth rate in height. The initial stage (0–100 s) has a smaller slope for dendritic crystals, suggesting more deposition in the side branches during this period. After 100 s, the slope gradually approaches that of needle-like crystals, indicating an accelerated tip growth rate. Fig. 3(d) displays the variation curve of the solid phase proportion (S_p) of multi-core growing lithium dendrites over time. Here, the solid phase proportion (S_p) is defined as the ratio of the volume occupied by the dendrite solid phase to the volume formed by the maximum dendrite height multiplied by the base area. The value of 1-S_p can be considered as the porosity during the multi-core dendrite growth process. In multi-core dendrite growth, large dendrites exhibit significant growth advantages, while smaller dendrites are suppressed, leading to a rapid decrease in the solid phase proportion. When the dendrites approach the upper surface, the rate of change stabilizes, and the solid phase proportion slightly increases.

3.2. The influence of anisotropy and temperature on the evolution of lithium dendrites

Based on the three-dimensional model, the temperature field was coupled to further explore the effects of anisotropy and temperature on the growth of lithium dendrites. We investigated the changes in maximum height and solid phase proportion of dendrites under different

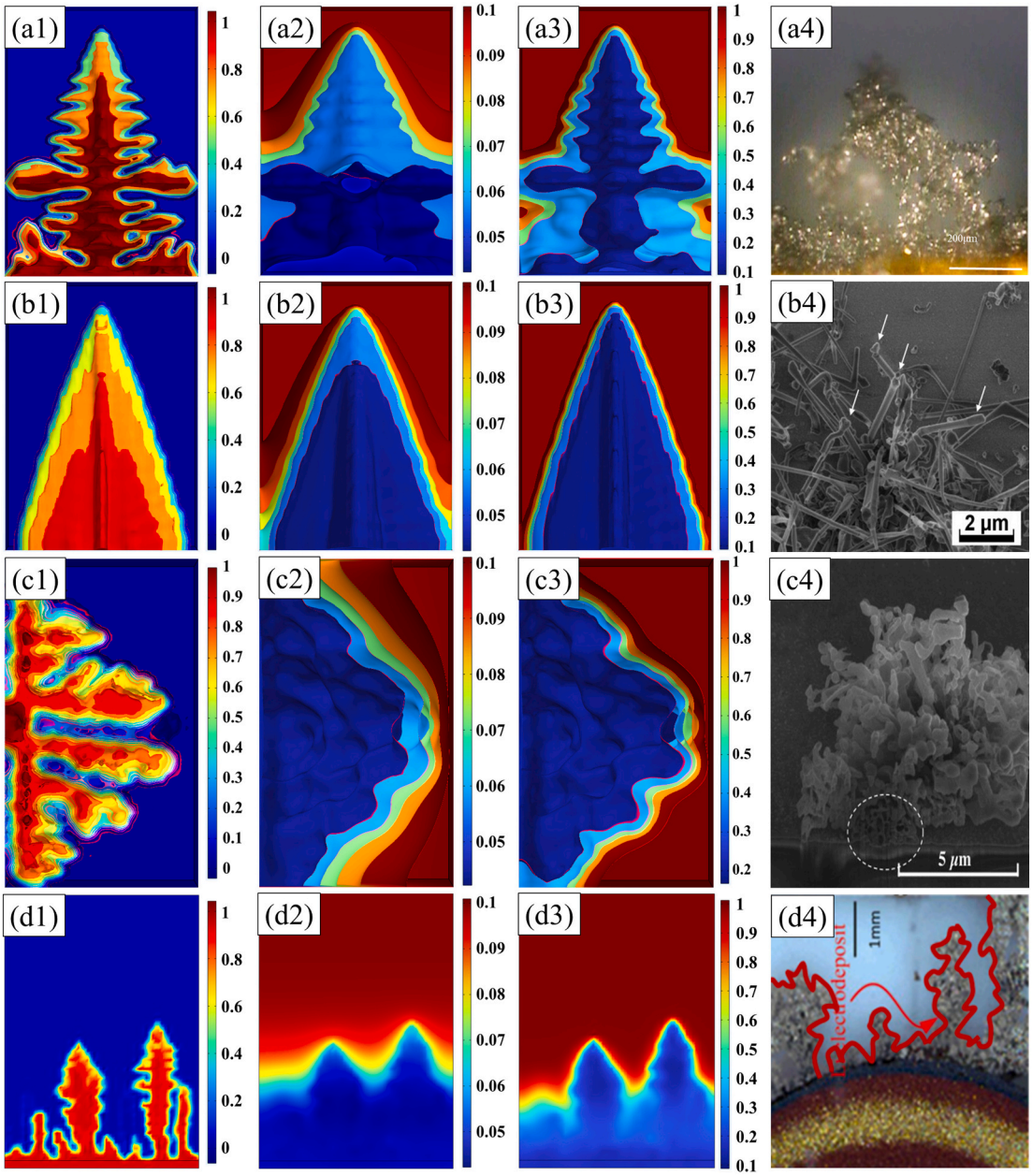
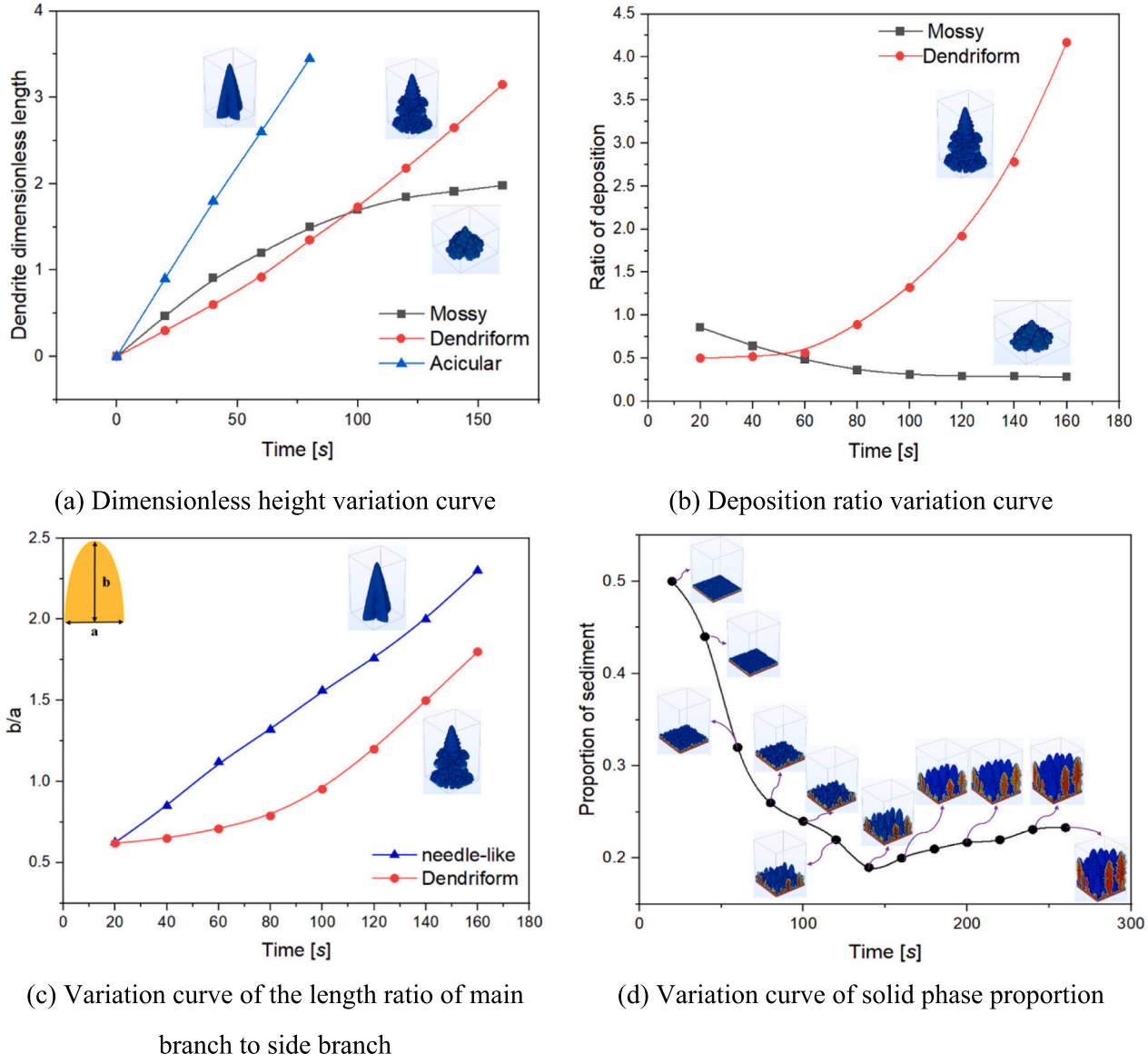


Fig. 2. Cross-sectional views of four types of three-dimensional lithium dendrite morphologies compared with experimental observations. ((a1)-(a4): dendritic lithium dendrites; (b1)-(b4): acicular lithium dendrites; (c1)-(c4): mossy lithium dendrites; (d1)-(d4): multinucleated growing lithium dendrites. The numbers 1, 2, and 3 represent cross-sectional plots of the phase fields, potentials, and concentrations, respectively, and 4 represents plots of the experimental results; more detailed 3D structural information is provided in Supporting Materials S1-S4.)



(a) Dimensionless height variation curve

(b) Deposition ratio variation curve

(c) Variation curve of the length ratio of main branch to side branch

(d) Variation curve of solid phase proportion

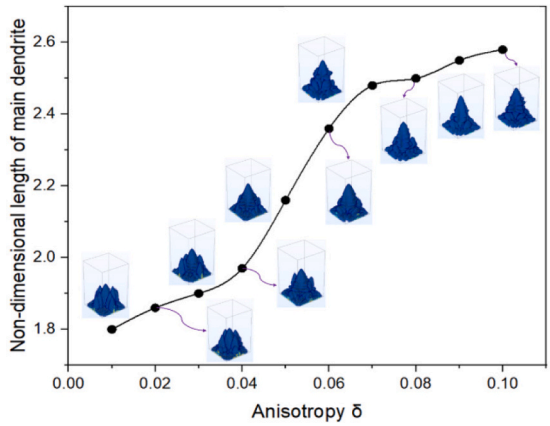
Fig. 3. Comparative analysis of growth curves of four types of lithium dendrites.

anisotropy and temperature conditions. At 160 s of dendrite growth, we analyzed the impact curves of different parameters on dendrite growth height and deposition ratio, as shown in Fig. 4. For detailed information on the influence of temperature and anisotropy on the three-dimensional structure of lithium dendrites, refer to Supporting Materials S5-S6.

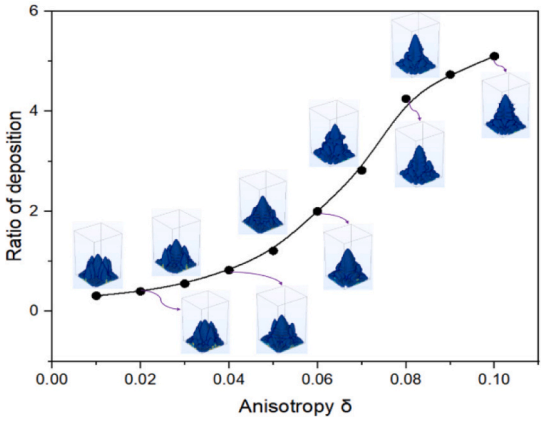
Fig. 4(a) shows the variation in dimensionless dendrite height with increasing anisotropy. Fig. 4(a) shows that the dendrite height increases rapidly as anisotropy changes from 0.04 to 0.07. During the change of anisotropy from 0.01 to 0.1, the dendrite height increases by 0.75. Overall, a decrease in anisotropy favors smooth dendrite deposition. Fig. 4(b) shows the variation in the deposition ratio of the main dendrite to side branches with increasing anisotropy. When anisotropy changes from 0.01 to 0.08, the curve is concave, indicating that an increase in anisotropy increases the deposition on the main dendrite while decreasing the deposition on the side branches. This combined effect leads to a rapid increase in the ratio, and after anisotropy reaches 0.08, the rate of increase in the deposition ratio slows down. Fig. 4(c) shows the relationship between dendrite height and temperature. It can be seen that as the temperature increases from 273 K to 343 K, the dendrite height decreases approximately linearly. The height drops by 1.3, accounting for 38 % of the initial height. After 343 K, the change tends to stabilize, indicating that ion diffusion is no longer the primary limiting factor for the reaction rate, and the electron transfer step controls the electrochemical reaction rate. Fig. 4(d) analyzes the relationship between the deposition ratio and temperature. The deposition ratio increases in the temperature range from 273 K to 293 K. Subsequently, in the temperature range from 293 K to 323 K, the deposition ratio decreases rapidly, resulting in the secondary dendrites on the main branch gradually shortening, fewer wrinkles, a smoother outer surface, and denser deposition. The fluctuation in the deposition ratio with temperature may be due to the combined effect of enhanced ion diffusion capacity and anisotropic dendrite growth.

3.3. The influence of potential distribution on dendrite evolution in multi-core lithium dendrites

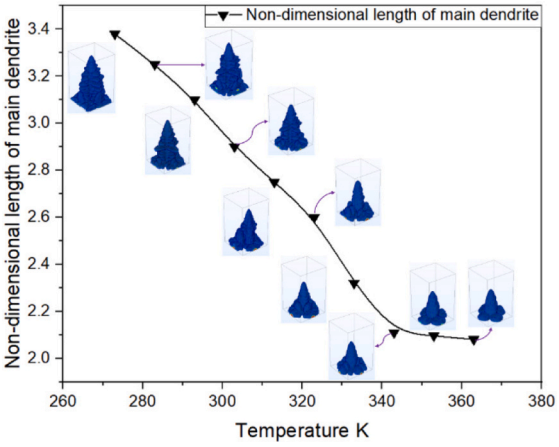
In the simulation study of three-dimensional multi-core lithium dendrite growth, it was found that the spatial potential field formed by conductive dendrites significantly impacted dendrite deposition. To further explore the interactions between dendrites, we performed a sectional analysis of three-dimensional multi-core lithium dendrites



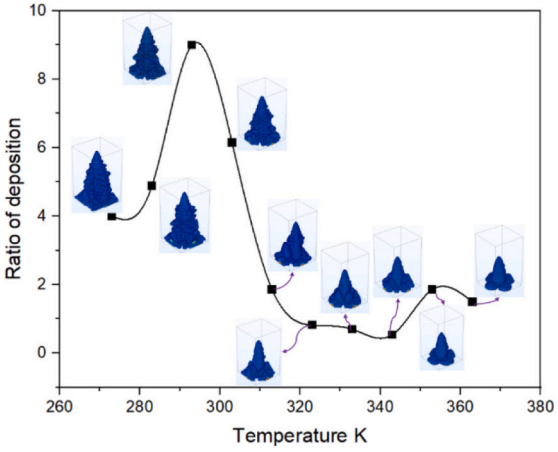
(a) The impact curve of anisotropy on dimensionless height



(b) The impact curve of anisotropy on the deposition ratio



(c) The impact curve of temperature on dimensionless height



(d) The impact curve of temperature on the deposition ratio

Fig. 4. The influence of temperature and anisotropy on dendrite evolution.

using cross-sections parallel to the *xoy* plane. We captured horizontal cross-sectional images (phase field, potential field, concentration field) at different heights at 280 s and horizontal cross-sectional images (phase field, potential field, concentration field) at different times at a cross-section of 0.25 height. Detailed cross-sectional images can be referred to in Supplementary Figures S_7-S_{12} and Video.

Contrary to the fusion view of dendrites, our analysis of the phase field cross-sectional images (S_7) revealed that dendrite growth is competitive. Larger lithium dendrites attract more lithium ions for deposition rather than forming through the fusion of smaller dendrites. From the cross-sectional image of the phase field at a height of 0.25 in Supplementary Figure S_7, it can be seen that after the formation of multi-core lithium dendrites, the bottom does not form a dense lithium deposit and remains as fine dendrites. This structure has poor stability, making it prone to breakage and the formation of dead lithium.

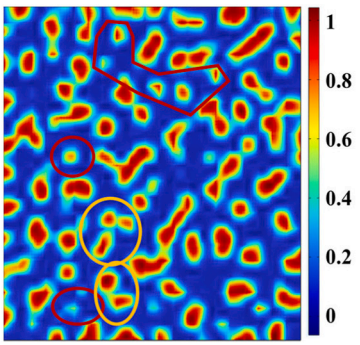
By analyzing the horizontal cross-sectional images of the phase field, concentration field, and potential field at a height of 0.25 at different times, we observe that from 140 s to 240 s, the dendrites within the red circle diminish, while the dendrites within the yellow circle tend to merge (Fig. 5 (a1) and (a2)). Further analysis of the concentration field and potential field (Fig. 5 (b1), (b2), (c1), and (c2)) leads us to conclude that as the lithium dendrites grow, the interaction between the spatial potential field formed by the conductive dendrites and the concentration gradient potential causes a redistribution of lithium deposition.

These findings suggest that by constructing a three-dimensional conductive structure in the anode, forming a three-dimensional potential field conducive to stable deposition, it is possible to achieve stable deposition and stripping of lithium metal. This research provides new insights into the growth mechanism of lithium dendrites and offers important references for designing more effective lithium metal deposition and stripping technologies.

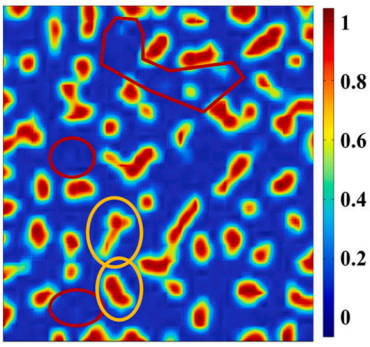
4. Conclusion

In this study, we developed a three-dimensional phase field model for lithium dendrite growth and employed Comsol software to simulate and quantitatively analyze the growth of various dendrite morphologies. Additionally, the temperature field was coupled to assess the effects of anisotropy and temperature on the physical parameters of dendrites. By analyzing the internal potential of multi-core lithium dendrites, we suggested a method to enhance stable lithium metal deposition.

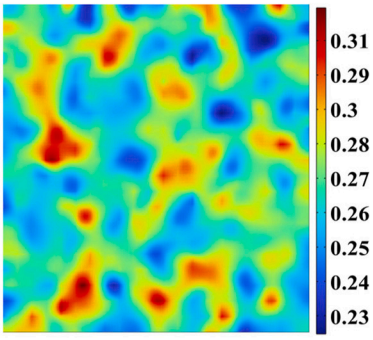
Our three-dimensional simulation results of multi-morphology lithium dendrites indicate that needle-like crystals exhibit the fastest growth in the vertical direction and their formation should be minimized. Mossy crystals are more likely to deposit around the periphery and maintain a lower height compared to dendritic crystals. During the



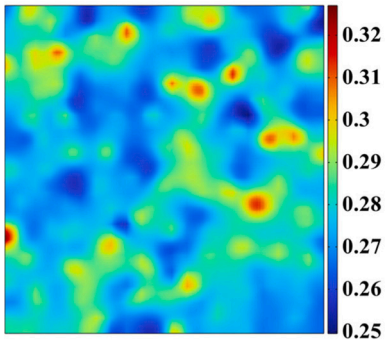
(a1) Horizontal cross-section of the phase field at height 0.25 at 140s.



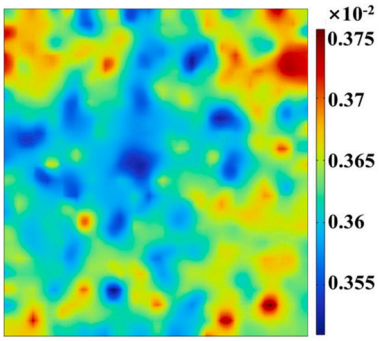
(a2) Horizontal cross-section of the phase field at height 0.25 at 240s.



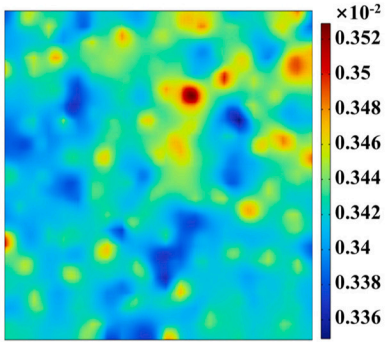
(b1) Horizontal cross-section of the concentration field at a height of 0.25 at 140s



(b2) Horizontal cross-section of the concentration field at a height of 0.25 at 240s



(c1) Horizontal cross-section image of the potential field at a height of 0.25 at 140s.



(c2) Horizontal cross-section image of the potential field at a height of 0.25 at 240s.

Fig. 5. Comparison of horizontal cross-section images of phase, concentration, and potential fields at a height of 0.25 at 140 and 240 s (Since the values of the physical parameters in the transverse cross-section are relatively close after the chemical reaction, in order to present the potential possible effects of potential and concentration on the redistribution of lithium deposition, we chose a smaller range of legends, with concentrations ranging from 0.23 to 0.32 and potentials ranging from 0.336 to 0.375.)

growth process, larger dendrites demonstrate a significant growth advantage, suppressing smaller dendrites and leading to a rapid decline in the solid phase ratio. As dendrites reach the opposite electrode, the rate of change stabilizes, and the solid phase ratio slightly increases. Our findings on temperature and anisotropy reveal that increased anisotropy enhances the maximum height of the dendrites, thus facilitating more lithium ion deposition on the main dendrite. Additionally, a higher temperature aids in the uniform deposition of lithium dendrites and effectively curtails their growth.

Further analysis of the internal potential in growing multi-core lithium dendrites showed that the spatial potential field created by conductive dendrites can lead to a redistribution of lithium deposition. Consequently, stable lithium metal deposition can be achieved by engineering a three-dimensional anode structure to produce an appropriate three-dimensional potential distribution. These insights offer valuable implications for understanding lithium dendrites' growth mechanisms and for designing more effective lithium metal deposition technologies.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.est.2024.113899.

Author statement

We promise that this paper will be original. We have addressed all

comments accordingly in both reply letter and revised manuscript, respectively. All changes made in revised manuscript have been highlighted in red. In addition, we have corrected some grammatical errors in the revised manuscript.

CRediT authorship contribution statement

Guoqing Qi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xunliang Liu:** Writing – review & editing, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Ruifeng Dou:** Writing – review & editing, Project administration, Methodology, Formal analysis, Conceptualization. **Zhi Wen:** Writing – review & editing, Software, Methodology, Formal analysis, Conceptualization. **Wenning Zhou:** Writing – review & editing, Software, Methodology, Formal analysis, Conceptualization. **Lin Liu:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by National Natural Science Foundation of China (No. 52476053; 52076012) and Beijing Natural Science Foundation (No. 3242017). Thanks to the support from the National Science Foundation under Grant Number 1840732. L.L. would like to thank the support from the Kansas Soybean Commission and KU General Research Fund.

References

[1] T. Krauskopf, F.H. Richter, W.G. Zeier, J. Janek, Physicochemical concepts of the Lithium metal anode in solid-state batteries, Chem. Rev. 120 (2020) 7745–7794, https://doi.org/10.1021/acs.chemrev.0c00431.
[2] X.-B. Cheng, R. Zhang, C.-Z. Zhao, Q. Zhang, Toward safe Lithium metal anode in rechargeable batteries: a review, Chem. Rev. 117 (2017) 10403–10473, https://doi.org/10.1021/acs.chemrev.7b00115.
[3] D. Lin, Y. Liu, Y. Cui, Reviving the Lithium metal anode for high-energy batteries, Nat. Nanotechnol. 12 (2017) 194–206, https://doi.org/10.1038/nnano.2017.16.
[4] W. Xu, J. Wang, F. Ding, X. Chen, E. Nasybulin, Y. Zhang, J.-G. Zhang, Lithium metal anodes for rechargeable batteries, Energy Environ. Sci. 7 (2014) 513–537, https://doi.org/10.1039/C3EE40795K.
[5] K. Kerman, A. Luntz, V. Viswanathan, Y.-M. Chiang, Z. Chen, Review—practical challenges hindering the development of solid state Li ion batteries, J. Electrochem. Soc. 164 (2017) A1731, https://doi.org/10.1149/2.1571707jes.
[6] C. Monroe, J. Newman, Dendrite growth in Lithium/polymer systems : a propagation model for liquid electrolytes under Galvanostatic conditions, J. Electrochem. Soc. 150 (2003) A1377, https://doi.org/10.1149/1.1606686.
[7] C. Monroe, J. Newman, The effect of interfacial deformation on electrodeposition kinetics, J. Electrochem. Soc. 151 (2004) A880, https://doi.org/10.1149/1.1710893.
[8] C. Monroe, J. Newman, The impact of elastic deformation on deposition kinetics at Lithium/polymer interfaces, J. Electrochem. Soc. 152 (2005) A396, https://doi.org/10.1149/1.1850854.
[9] C. Monroe, J. Newman, Dendrite growth in Lithium/polymer systems : a propagation model for liquid electrolytes under Galvanostatic conditions, J. Electrochem. Soc. 150 (2003) A1377, https://doi.org/10.1149/1.1606686.
[10] L.-Q. Chen, Phase-field models for microstructure evolution, Annu. Rev. Mater. Res. 32 (2002) 113–140, https://doi.org/10.1146/annurev.matsci.32.112001.132041.
[11] L.-Q. Chen, Y. Zhao, From classical thermodynamics to phase-field method, Prog. Mater. Sci. 124 (2022) 100868, https://doi.org/10.1016/j.pmatsci.2021.100868.
[12] L. Chen, H.W. Zhang, L.Y. Liang, Z. Liu, Y. Qi, P. Lu, J. Chen, L.-Q. Chen, Modulation of dendritic patterns during electrodeposition: a nonlinear phase-field model, J. Power Sources 300 (2015) 376–385, https://doi.org/10.1016/j.jpowsour.2015.09.055.
[13] Z. Hong, V. Viswanathan, Phase-field simulations of Lithium dendrite growth with open-source software, ACS Energy Lett. 3 (2018) 1737–1743, https://doi.org/10.1021/acsenergylett.8b01009.
[14] H.H. Yan, Y.H. Bie, X.Y. Cui, G.P. Xiong, L. Chen, A computational investigation of thermal effect on Lithium dendrite growth, Energy Convers. Manag. 161 (2018) 193–204, https://doi.org/10.1016/j.enconman.2018.02.002.
[15] Z. Hong, V. Viswanathan, Prospect of thermal shock induced healing of Lithium dendrite, ACS Energy Lett. 4 (2019) 1012–1019, https://doi.org/10.1021/acsenergylett.9b00433.
[16] W. Mu, X. Liu, Z. Wen, L. Liu, Numerical simulation of the factors affecting the growth of Lithium dendrites, J. Energy Storage 26 (2019) 100921, https://doi.org/10.1016/j.est.2019.100921.
[17] D. Qiao, X. Liu, R. Dou, Z. Wen, W. Zhou, L. Liu, Quantitative analysis of the inhibition effect of rising temperature and pulse charging on Lithium dendrite growth, J. Energy Storage 49 (2022) 104137, https://doi.org/10.1016/j.est.2022.104137.
[18] L. Porz, T. Swamy, B.W. Sheldon, D. Rettenwander, T. Frömling, H.L. Thaman, S. Berendts, R. Uecker, W.C. Carter, Y.-M. Chiang, Mechanism of Lithium metal penetration through inorganic solid electrolytes, Adv. Energy Mater. 7 (2017) 1701003, https://doi.org/10.1002/aenm.201701003.
[19] K. Tantratian, H. Yan, K. Ellwood, E.T. Harrison, L. Chen, Unraveling the Li penetration mechanism in polycrystalline solid electrolytes, Adv. Energy Mater. 11 (2021) 2003417, https://doi.org/10.1002/aenm.202003417.
[20] C. Yuan, X. Gao, Y. Jia, W. Zhang, Q. Wu, J. Xu, Coupled crack propagation and dendrite growth in solid electrolyte of all-solid-state battery, Nano Energy 86 (2021) 106057, https://doi.org/10.1016/j.nanoen.2021.106057.
[21] C. Yuan, W. Lu, J. Xu, Unlocking the Electrochemical–Mechanical Coupling Behaviors of Dendrite Growth and Crack Propagation in All-Solid-State Batteries, Adv. Energy Mater. 11 (2021) 2101807. doi:https://doi.org/10.1002/aenm.202101807.
[22] Z. Ning, D.S. Jolly, G. Li, R. De Meyere, S.D. Pu, Y. Chen, J. Kasemchainan, J. Ihli, C. Gong, B. Liu, D.L.R. Melvin, A. Bonnin, O. Magdysyuk, P. Adamson, G. O. Hartley, C.W. Monroe, T.J. Marrow, P.G. Bruce, Visualizing plating-induced cracking in Lithium-anode solid-electrolyte cells, Nat. Mater. 20 (2021) 1121–1129, https://doi.org/10.1038/s41563-021-00967-8.
[23] P. Guan, L. Liu, Phase-field simulation of Lithium ion diffusion in solid electrolyte interphase, ECS Trans. 66 (2015) 81, https://doi.org/10.1149/06609.0081ecst.
[24] L. Liu, P. Guan, Phase-field modeling of solid electrolyte interphase (SEI) evolution: considering cracking and dissolution during battery cycling, ECS Trans. 89 (2019) 101, https://doi.org/10.1149/08901.0101ecst.
[25] P. Guan, L. Liu, Y. Gao, Phase-field modeling of solid electrolyte interphase (SEI) cracking in Lithium batteries, ECS Trans. 85 (2018) 1041, https://doi.org/10.1149/08513.1041ecst.
[26] H. Chen, A. Pei, J. Wan, D. Lin, R. Vilá, H. Wang, D. Mackanic, H.-G. Steinrück, W. Huang, Y. Li, A. Yang, J. Xie, Y. Wu, H. Wang, Y. Cui, Tortuosity effects in Lithium-metal host anodes, Joule 4 (2020) 938–952, https://doi.org/10.1016/j.joule.2020.03.008.
[27] R. Zhang, X. Shen, X.-B. Cheng, Q. Zhang, The dendrite growth in 3D structured lithium metal anodes: Electron or ion transfer limitation? Energy Storage Mater. 23 (2019) 556–565, https://doi.org/10.1016/j.ensm.2019.03.029.
[28] R. Zhang, X. Shen, H.-T. Ju, J.-D. Zhang, Y.-T. Zhang, J.-Q. Huang, Driving Lithium to deposit inside structured Lithium metal anodes: a phase field model, J. Energy Chem. 73 (2022) 285–291, https://doi.org/10.1016/j.jechem.2022.06.010.
[29] C. Zhang, S. Zhu, D. Wang, C. Lei, Y. Zhao, Phase field modeling of Lithium deposition in porous Lithium metal anodes, J. Energy Storage 73 (2023) 109279, https://doi.org/10.1016/j.est.2023.109279.
[30] D. Bistri, A. Afshar, C.V. Di Leo, Modeling the chemo-mechanical behavior of all-solid-state batteries: a review, Meccanica 56 (2021) 1523–1554, https://doi.org/10.1007/s11012-020-01209-y.
[31] C.-P. Yang, Y.-X. Yin, S.-F. Zhang, N.-W. Li, Y.-G. Guo, Accommodating Lithium into 3D current collectors with a submicron skeleton towards long-life Lithium metal anodes, Nat. Commun. 6 (2015) 8058, https://doi.org/10.1038/ncomms9058.
[32] J. Yun, B.-K. Park, E.-S. Won, S.H. Choi, H.C. Kang, J.H. Kim, M.-S. Park, J.-W. Lee, Bottom-up Lithium growth triggered by interfacial activity gradient on porous framework for Lithium-metal anode, ACS Energy Lett. 5 (2020) 3108–3114, https://doi.org/10.1021/acsenergylett.0c01619.
[33] L.-L. Lu, J. Ge, J.-N. Yang, S.-M. Chen, H.-B. Yao, F. Zhou, S.-H. Yu, Free-standing copper nanowire network current collector for improving Lithium anode performance, Nano Lett. 16 (2016) 4431–4437, https://doi.org/10.1021/acs.nanolett.6b01581.
[34] J. Li, P. Zou, S.W. Chiang, W. Yao, Y. Wang, P. Liu, C. Liang, F. Kang, C. Yang, A conductive-dielectric gradient framework for stable Lithium metal anode, Energy Storage Mater. 24 (2020) 700–706, https://doi.org/10.1016/j.ensm.2019.06.019.
[35] S.Y. Han, J.A. Lewis, P.P. Shetty, J. Tippens, D. Yeh, T.S. Marchese, M. T. McDowell, Porous metals from chemical Dealloying for solid-state battery anodes, Chem. Mater. 32 (2020) 2461–2469, https://doi.org/10.1021/acs.chemmater.9b04992.
[36] Z. Shen, W. Zhang, S. Li, S. Mao, X. Wang, F. Chen, Y. Lu, Tuning the interfacial electronic conductivity by artificial Electron tunneling barriers for practical Lithium metal batteries, Nano Lett. 20 (2020) 6606–6613, https://doi.org/10.1021/acs.nanolett.0c02371.
[37] Y. Liu, X. Xu, O.O. Kapitanova, P.V. Evdokimov, Z. Song, A. Matic, S. Xiong, Electro-chemo-mechanical modeling of artificial solid electrolyte interphase to

enable uniform electrodeposition of Lithium metal anodes, Adv. Energy Mater. 12 (2022) 2103589, https://doi.org/10.1002/aenm.202103589.
[38] W. Zhang, Z. Shen, S. Li, L. Fan, X. Wang, F. Chen, X. Zang, T. Wu, F. Ma, Y. Lu, Engineering wavy-nanostructured anode interphases with fast ion transfer kinetics: toward practical Li-metal full batteries, Adv. Funct. Mater. 30 (2020) 2003800, https://doi.org/10.1002/adfm.202003800.
[39] S. Li, S.-Q. Zhang, L. Shen, Q. Liu, J.-B. Ma, W. Lv, Y.-B. He, Q.-H. Yang, Progress and perspective of ceramic/polymer composite solid electrolytes for Lithium batteries, Adv. Sci. 7 (2020) 1903088, https://doi.org/10.1002/advs.201903088.
[40] S. Xu, D.W. McOwen, C. Wang, L. Zhang, W. Luo, C. Chen, Y. Li, Y. Gong, J. Dai, Y. Kuang, C. Yang, T.R. Hamann, E.D. Wachsman, L. Hu, Three-Dimensional, Solid-State Mixed Electron–Ion Conductive Framework for Lithium Metal Anode, Nano Lett. 18 (2018) 3926–3933. doi:https://doi.org/10.1021/acs.nanolett.8b01295.
[41] Z. Mu, Z. Guo, Y.-H. Lin, Simulation of 3-D Lithium dendritic evolution under multiple electrochemical states: a parallel phase field approach, Energy Storage Mater. 30 (2020) 52–58, https://doi.org/10.1016/j.ensm.2020.04.011.
[42] M.E. Arguello, N.A. Labanda, V.M. Calo, M. Gumulya, R. Utikar, J. Derksen, Dendrite formation in rechargeable Lithium-metal batteries: phase-field modeling using open-source finite element library, J. Energy Storage 53 (2022) 104892, https://doi.org/10.1016/j.est.2022.104892.
[43] M.E. Arguello, N.A. Labanda, V.M. Calo, M. Gumulya, R. Utikar, J. Derksen, Three-dimensional experimental-scale phase-field modeling of dendrite formation in rechargeable Lithium-metal batteries, J. Energy Storage 62 (2023) 106854, https://doi.org/10.1016/j.est.2023.106854.
[44] R. Acharya, J.A. Sharon, A. Staroselsky, Prediction of microstructure in laser powder bed fusion process, Acta Mater. 124 (2017) 360–371, https://doi.org/10.1016/j.actamat.2016.11.018.
[45] M.S. Park, S.B. Ma, D.J. Lee, D. Im, S.-G. Doo, O. Yamamoto, A highly reversible Lithium metal anode, Sci. Rep. 4 (2014) 3815, https://doi.org/10.1038/srep03815.
[46] J. Steiger, D. Kramer, R. Mönig, Microscopic observations of the formation, growth and shrinkage of Lithium Moss during electrodeposition and dissolution, Electrochim. Acta 136 (2014) 529–536, https://doi.org/10.1016/j.electacta.2014.05.120.
[47] J. Steiger, D. Kramer, R. Mönig, Mechanisms of dendritic growth investigated by in situ light microscopy during electrodeposition and dissolution of Lithium, J. Power Sources 261 (2014) 112–119, https://doi.org/10.1016/j.jpowsour.2014.03.029.
[48] S.J. Harris, A. Timmons, D.R. Baker, C. Monroe, Direct in situ measurements of Li transport in Li-ion battery negative electrodes, Chem. Phys. Lett. 485 (2010) 265–274, https://doi.org/10.1016/j.cplett.2009.12.033.
[49] H.-W. Zhang, Z. Liu, L. Liang, L. Chen, Y. Qi, S.J. Harris, P. Lu, L.-Q. Chen, Understanding and predicting the Lithium dendrite formation in Li-ion batteries: phase field model, ECS Trans. 61 (2014) 1–9, https://doi.org/10.1149/06108.0001ecst.
[50] W. Jiang, Q. Zhou, F. Lu, Y. Chen, Z. Ma, A thermal-electrochemical-mechanical coupled model based on non-equilibrium thermodynamics of Li-ion batteries, J. Energy Storage 55 (2022) 105655, https://doi.org/10.1016/j.est.2022.105655.
[51] W. Jiang, Z. Wang, L. Hu, Y. Wang, Z. Ma, Simulations of dendrite and crack and their interactions in solid electrolyte by phase field method, J. Energy Storage 86 (2024) 111126, https://doi.org/10.1016/j.est.2024.111126.